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Angular-momentum resolution in transitional-mode state counting for loose transition states

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The classical evaluation of the angular-momentum-resolved sum of states for the loosely hindered rotational degrees of freedom, i.e., the transitional modes, in loose transition states occurring in unimolecular dissociation, radical-radical recombination, ion-molecule, and other collision-complex-forming bimolecular reactions is considered. Exact analytic expressions are derived for the momentum-space volume available to the transitional modes at a given configuration $\mathbf{q}$ with energy E and total angular momentum vector $\mathbf{J}$. The results are completely general with respect to the type of fragment rotors involved and their relative orientation within the loose transition state, and constitute a dramatically simplified technique for J -resolved classical state counting. The utility of the expressions lies in the fact that they obviate the necessity of numerical integration over the system's momentum space, thus reducing substantially the computational effort involved in the exact evaluation of the transitional-mode sum of states. The present results verify expressions which were postulated to apply to arbitrary configurations in our earlier work.

I. INTRODUCTION

There is a continuing drive on the part of theoretical and experimental kineticists to measure, model, understand, and finally predict the rates of reactions proceeding on potential surfaces with no significant barrier to recombination (e.g., Refs. 1–6). Such reactions are common and often display a negative temperature dependence.² They are also challenging and difficult to model because of the marked variation of the transition state with energy and angular momentum, the wide range of contributing angular momenta, and the multidimensional, nonseparable nature of the transitional modes (i.e., those which transform from fragment and orbital rotations at large separations to vibrations and overall rotation in the unimolecular species).

A number of different methods may be applied to calculate rate coefficients for these reactions. Close-coupled quantum scattering calculations (e.g., Refs. 7–9) and classical trajectory calculations have been carried out for some systems (e.g., Refs. 10–12). However, such calculations are computationally too demanding for application on a routine basis and they require more information concerning the potential surface than is commonly available (unless one restricts oneself to the long-range dynamics where electrostatic forces dominate).^{7,13–16} The need for accurate calculations of potential surfaces in such reactions is gradually being answered^{17–21} and quality surfaces are beginning to appear. There is a corresponding need, however, for the development of faster methods which can be used to calculate accurately rate coefficients on a more routine basis. Of the faster methods, statistical theories are the most commonly used and the most systematically developed.^{22,23} Statistical theories for simple-fission unimolecular dissociation and for recombination or chemical activation without a barrier in the entrance channel have generally been either adiabatic (e.g., Refs. 22 and 24–26) or variational (e.g.,

Refs. 4, 27, and 28) in nature with regard to the treatment of the loose (i.e., flexible) transition state. The statistical expression for the microscopic rate coefficient $k(E, J)$ is^{23,29,30}

$$k(E, J) = \frac{W(E, J)}{h\rho(E, J)}, \quad (1)$$

where h is Planck's constant and $W(E, J)$ is the number of open channels within the context of the adiabatic approach or the sum of states at the variationally optimized transition state within the context of variational transition state theory. The density of states $\rho(E, J)$ will be that for the bimolecular reactants in the case of recombination, or that of the unimolecular reactant in the case of unimolecular dissociation. Recently, experimental evidence concerning product state distributions in unimolecular dissociation reactions^{31–35} has led to the conclusion that a judicious mixture of the two approaches, in which vibrations are treated adiabatically and hindered rotations variationally, may be more appropriate than either limiting case.^{36–39} A similar conclusion was reached independently in the application of variational statistical theory to the calculation of ion-dipole association rates.⁴⁰

The complexity of the noncentral interaction potential and the coupling between the angular momenta of the fragments and the orbital rotation in loose transition states makes a rigorous evaluation of the eigenstate patterns in the region of intermediate separation an extremely difficult task, even for relatively simple systems.^{13,41} In recent years, classical state counting methods have been utilized to carry out rigorous statistical calculations of dissociation and recombination reaction rate coefficients with full account of these complexities, insofar as the knowledge of the potential surface and the classical approximation allow.^{4,20,37–39,42–46} In the classical approximation to the sum of states, the volume in phase space (in units of Planck's

constant h) is determined which contains all classically allowed configuration and momentum states $(\mathbf{q}, \mathbf{p})$, subject to energy and angular momentum constraints. Thus, the sum of states for a given energy E and angular momentum J is written

$$W(E, J) = h^{-n} \int \cdots \int d\mathbf{q} d\mathbf{p} S(E - H) \delta(J - j), \quad (2)$$

where n is the relevant number of degrees of freedom $d\mathbf{q} = dq_1 \cdots dq_n$, $d\mathbf{p} = dp_1 \cdots dp_n$. H the Hamiltonian and j the system angular momentum are functions of $\mathbf{q}$ and $\mathbf{p}$. The Dirac delta function imposes the constraint that j take the specified value J . The Heaviside step function imposes the constraint that the system energy H must not exceed the specified maximum E ; $S(E - H) = 0$ for $H > E$ and $S(E - H) = 1$ for $H < E$.

The classical approximation to the sum of states is expected to be most accurate in application to those degrees of freedom which are not hindered by a strong restoring force to motion about the minimum-energy configuration and hence have a relatively small spacing between quantum states. Degrees of freedom which fall into this category are typically free or loosely hindered rotations and, of course, translational motion. Once the sum of states for the “loose” degrees of freedom has been determined classically, it is convoluted easily with the quantized state pattern for the vibrational degrees of freedom by direct count methods.^{23,47-49} The high dimensionality of the phase-space integrals involved makes the evaluation of sums of states for the multidimensional, typically nonseparable transitional modes over the range of E and J required for the calculation of a thermal rate coefficient a demanding numerical task. It is well known that when angular-momentum constraints are not imposed, the evaluation of densities of states, sums of states, and partition functions may be simplified considerably by analytic evaluation of the momentum integrals.⁵⁰⁻⁵³ This fact has been utilized in several recent applications of microcanonical variational transition state theory (μ VTST).^{46,53-55} Although high-pressure-limiting recombination rates (sometimes termed *capture* rates) can often be calculated quite accurately by models which carry out E -resolved optimization of the transition state [i.e., variational calculation of $k(E)$],^{53,54,56} as opposed to the more accurate E - and J -resolved optimization [i.e., variational calculation of $k(E, J)$], angular momentum conservation becomes a crucial factor in pressure-dependent calculations.^{57,58} This is particularly so when additional decomposition channels with “tight” transition states (i.e., transition states situated at the top of a barrier) are involved.⁵⁹⁻⁶² In such situations, the calculation of microscopic rate coefficients $k(E, J)$ becomes essential. The complexity of the constraint imposed on phase-space integrals by a specified angular momentum has hitherto been dealt with by numerical integration over the complete dimensionality of the phase space^{4,37-39,42-46,54} (minus the conserved quantities J and its projection onto an arbitrary space-fixed axis M , and their conjugate angles).

In a recent analysis, we have presented approximate analytical formulas for the momentum-space volume available to the transitional modes $\Phi(E, J, \mathbf{q})$ at a given angular configuration $\mathbf{q}$ of the fragments.⁶³ Use of these formulas obviates the necessity of numerical integration over the full dimensionality of the transitional-mode phase space, reducing the task to a numerical integration over only those internal angular coordinates which appear in the potential. The magnitude of the computational task involved in such variational calculations is thus decreased substantially. The analytic expressions derived take complete account of the inertial symmetry (or asymmetry) of the individual fragments, requiring only that the *overall* body of the loose transition state be accurately approximated as a symmetric top (generally an excellent approximation due to the highly prolate nature of simple-fission-type transition states, wherein one bond has been greatly stretched). This result was obtained by consideration of a specific set of configurations $\mathbf{q}_i$ in which the principal axes of the two fragments and of the orbital rotation are coaligned. For such configurations, the absence of angular coupling in the expressions for the total system angular momentum and the kinetic energy indicated intuitively obvious “center-of-mass” and “skewed-coordinate” type momentum transformations, which led to analytical expressions for $\Phi(E, J, \mathbf{q}_i)$.⁶³ The observation that, for each of these “reference” configurations $\mathbf{q}_i$, the analytic expressions for the momentum space obtained conform to a generic functional form led to the suggestion that this function should also apply accurately for intermediate (arbitrary) configurations $\mathbf{q}$.

In the present study, we generalize the considerations of the previous work to analyze the momentum-space volume available to the transitional modes for arbitrary configurations $\mathbf{q}$. Exact analytic expressions are derived for the available momentum-space volume at an arbitrary configuration of a system with a given total energy E and total angular momentum vector $\mathbf{J}$. These expressions are completely general with regard to the type of fragments involved in the loose transition state. Integration over the different possible body-fixed orientations of the $\mathbf{J}$ vector requires a one-dimensional numerical integral for exact results, but may be carried out analytically if the *overall* body is approximated as a symmetric top. Approximating the loose transition state at a given configuration $\mathbf{q}$ as a prolate top and using the resulting analytical formula for the momentum-space volume is found to generate exceedingly accurate results. The expressions derived presently for $\Phi(E, J, \mathbf{q})$ turn out to be precisely those postulated in our previous work,⁶³ confirming the suspicion, based on numerical tests, that these expressions worked too well to be approximate.

II. MOMENTUM-SPACE VOLUME FOR THE J -RESOLVED TRANSITIONAL-MODE SUM OF STATES

In recombining or dissociating systems wherein there is no significant barrier to recombination, a “loose” transition state describes a rate-determining position along the reaction coordinate which involves two partially separated

fragments which are essentially independent with regard to their internal vibrations, but still interact through their rotations because of the angular dependence of the interaction potential. Since the vibrations of the fragments contribute negligibly to the overall angular momentum in comparison with the rotational degrees of freedom, separability is commonly assumed for the purposes of state counting so that the vibrational states are counted directly subject only to energetic constraints, and the rotational states are counted classically subject to energy and angular momentum constraints. The vibrational density of states $\rho_{\text{vib}}(E)$ and transitional-mode sum of states $W_{\text{TM}}(E, J)$ may be convoluted together to generate the total sum of states in the loose transition state

$$W(E, J) = \int_{E_{\min}(J)}^{E^*} W_{\text{TM}}(E_+, J) \rho_{\text{vib}}(E^* - E_+) dE_+, \quad (3)$$

where $E_{\min}(J)$ is the minimum energy required to generate the angular momentum J and

$$E^* = E - V(R). \quad (4)$$

In Eq. (4), $V(R)$ is the potential of the system in its minimum-energy configuration at the given separation R . Equivalently, $W(E, J)$ may be obtained by convoluting the vibrational *sum* of states with the transitional-mode *density* of states (e.g., Ref. 4).

The vibrational state densities are easily evaluated by standard direct count techniques (e.g., Ref. 23). The evaluation of the state pattern for the transitional modes is considerably more complicated due to the nonseparable nature of the interaction potential and the complexity of the angular momentum coupling between the two fragments and the orbital rotation. The case where the fragments are able to rotate freely under a central interaction potential is treated by phase space theory (PST).⁶⁴⁻⁶⁸ However, the noncentrality of the interaction potential in most reactions of this sort requires a departure from the free-rotor model, since the variationally determined transition states typically lie at separations where the fragment rotations are hindered significantly.^{42,45,54} Hindrance to the rotational motion of the fragments can be accounted for with various levels of approximation (e.g., Refs. 4-6 and 23). Simple model potentials which mimic the most important features of the interaction potential have been utilized by a number of authors.^{42,69-73} Such models are usually implemented at the level of canonical variation of the transition state, as opposed to the more accurate microcanonical variation, and generally involve angular-momentum approximations equivalent to the "pseudodiatomic," or "sudden" methods.^{14,15,23} Analytic approximations to the sum of states which involve better treatment of angular momentum are beginning to emerge.⁶³ A more complete treatment of the interaction potential requires numerical evaluation of the necessary phase space integral for the sum of states, as typified by the pioneering work of Wardlaw and Marcus.^{4,42} As indicated in the Introduction, our purpose in this paper is to show that integration over the full dimensionality of the phase

space is no longer necessary since the momentum integrals can be evaluated analytically.

The classical approximation to the sum of states for the transitional modes may be written

$$W_{\text{TM}}(E, J) = \frac{1}{\sigma_1 \sigma_2} \int d\Gamma S(E - H_{\text{TM}}) \delta(J - j), \quad (5)$$

where $d\Gamma = d\mathbf{q} d\mathbf{p}/h^n$ (n being the number of transitional modes) if conventional Euler angles and momenta are used⁴⁴ and $d\Gamma = d\mathbf{p} d\mathbf{q}/(2\pi)^n$ if action-angle variables are used,⁴² σ_1 and σ_2 are the symmetry numbers of the fragments, and H_{TM} is the Hamiltonian for the transitional modes.

The following derivation will be carried out for the most general case of a loose transition state consisting of two asymmetric top fragments: The requisite formulas for the simpler cases of atomic, linear, spherical-top, or symmetric-top fragments are obtained easily by elimination of degrees of freedom or equating the appropriate rotational constants as necessary. The reaction coordinate in the present work is chosen as the center-of-mass separation R . Alternative reaction coordinates, e.g., the length of the breaking or forming bond, have been considered by other authors.^{45,46} Our point of departure is the conventional Euler angles and their conjugate momenta $\{\theta_i, \phi_i, \psi_i, p_{\theta_i}, p_{\phi_i}, p_{\psi_i}\}$ for the fragments and $\{\theta, \phi, p_{\theta}, p_{\phi}\}$ for the orbital rotation. Transforming from the Euler momenta to the principal-axis momenta of the fragments and of the orbital rotation $\{j_1, j_2, l\}$,⁶³ Eq. (5) becomes

$$\begin{aligned} W_{\text{TM}}(E, J) = & \frac{1}{(2\pi)^8 \sigma_1 \sigma_2} \\ & \times \int \cdots \int d\theta_1 d\phi_1 d\psi_1 d\theta_2 d\phi_2 d\psi_2 d\theta d\phi \\ & \times \sin \theta_1 \sin \theta_2 \sin \theta d\mathbf{j}_1 d\mathbf{j}_2 d\mathbf{l} \\ & \times S(E^* - H_{\text{TM}}) \delta(J - j), \end{aligned} \quad (6)$$

where the principal-axis momenta are written as dimensionless quantities in units of $\hbar$ and $d\mathbf{j}_i = dj_{ix} dj_{iy} dj_{iz}$ and $d\mathbf{l} = dl_x dl_y dl_z$. In terms of the principal-axis momenta, the Hamiltonian H_{TM} is written

$$H_{\text{TM}} = T_{\text{rot}} + V(\mathbf{q}), \quad (7)$$

where

$$\begin{aligned} T_{\text{rot}} = & A_1 j_{1z}^2 + B_1 j_{1y}^2 + C_1 j_{1x}^2 + A_2 j_{2z}^2 + B_2 j_{2y}^2 + C_2 j_{2x}^2 \\ & + B_0 (l_x^2 + l_y^2) \end{aligned} \quad (8)$$

and

$$V(\mathbf{q}) = V(R, \mathbf{q}) - V(R). \quad (9)$$

In Eq. (8), A_i , B_i , and C_i are the rotational constants of fragment i and B_0 is that of the orbital rotation (i.e., $\hbar^2/2\mu R^2$). In Eq. (9), $\mathbf{q}$ represents the angular coordinates and $V(R)$ is the minimum potential at center-of-mass separation R as in Eq. (4). The available momentum space $\Phi(E, J, \mathbf{q})$ is defined as⁶³

$$\Phi(E, J, \mathbf{q}) = \frac{1}{\pi^3} \int \cdots \int d\mathbf{j}_1 d\mathbf{j}_2 d\mathbf{l} S(E^* - H_{\text{TM}}) \delta(J - j) \quad (10)$$

such that

$$W_{\text{TM}}(E, J) = \frac{1}{\sigma_1 \sigma_2 N} \int \cdots \int d\theta_1 d\phi_1 d\psi_1 d\theta_2 d\phi_2 d\psi_2 \times d\theta d\phi \sin \theta_1 \sin \theta_2 \sin \theta \Phi(E, J, \mathbf{q}), \quad (11)$$

where $N = 2^8 \pi^5$ is the normalizing factor for the angular integrals. It is useful to define a momentum space $\Phi(E, \mathbf{J}, \mathbf{q})$ for a given energy E and angular momentum vector $\mathbf{J}$,

$$\Phi(E, \mathbf{J}, \mathbf{q}) = \frac{1}{\pi^3} \int \cdots \int d\mathbf{j}_1 d\mathbf{j}_2 d\mathbf{l} S(E^* - H_{\text{TM}}) \delta(\mathbf{J} - \mathbf{j}) \quad (12)$$

such that

$$\Phi(E, J, \mathbf{q}) = \int d\mathbf{j} \Phi(E, \mathbf{j}, \mathbf{q}) \delta(J - j) \quad (13)$$

Our aim is to transform from $\{\mathbf{j}_1, \mathbf{j}_2, \mathbf{l}\}$ to a new set of "internal" angular momentum vectors $\{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_0\}$ with the following properties: (a) they do not contribute to the external angular momentum vector $\mathbf{J}$; and (b) they do not contribute to the external rotational energy E^J i.e., the transitional-mode kinetic energy expression (8) will separate into external rotational and internal rotational parts with no Coriolis coupling terms. Such a transformation will enable the analytic evaluation of the momentum-space volume $\Phi(E, \mathbf{J}, \mathbf{q})$. The method of achieving this separation derives from an ingenious technique developed by Jellinek and Li⁷⁴ for separating instantaneously (i.e., at a given configuration $\mathbf{q}$) the kinetic energy expression for an arbitrary N -body system into contributions from overall rotation, which are dictated by the total angular momentum vector $\mathbf{J}$, and contributions from internal momentum vectors, which by definition make no net contribution to the overall angular momentum. To obtain the necessary internal angular momenta, consider a transformation of the form

$$\mathbf{j}_1 = \mathbf{j}_1^J + \mathbf{p}_1, \quad \mathbf{j}_2 = \mathbf{j}_2^J + \mathbf{p}_2, \quad \mathbf{l} = \mathbf{l}^J + \mathbf{p}_0. \quad (14)$$

In Eq. (14), the vectors $\{\mathbf{j}_1, \mathbf{j}_1^J, \mathbf{p}_1\}$ are expressed in terms of the principal-axial system of fragment 1, $\{\mathbf{j}_2, \mathbf{j}_2^J, \mathbf{p}_2\}$ are expressed in terms of the principal-axial system of fragment 2, and $\{\mathbf{l}, \mathbf{l}^J, \mathbf{p}_0\}$ are expressed in terms of the principal-axial system of the orbital rotation. We shall require that $\mathbf{j}_1^J$, $\mathbf{j}_2^J$, and $\mathbf{l}^J$ be vector constants, to be determined below, which add to generate the total net angular momentum $\mathbf{J}$. To write this condition formally, it is necessary first to project the vectors onto a common set of cartesian axes. Nominating for convenience the principal axes of the orbital rotation as the common Cartesian axes, the vectors

projected onto this axial system are denoted with a prime. The condition is therefore written

$$\mathbf{j}_1^{J'} + \mathbf{j}_2^{J'} + \mathbf{l}^{J'} = \mathbf{J}' \quad (15)$$

hence

$$\sum_{i=0}^2 \mathbf{p}_i' = \mathbf{j}_1' + \mathbf{j}_2' + \mathbf{l}' - (\mathbf{j}_1^{J'} + \mathbf{j}_2^{J'} + \mathbf{l}^{J'}) = \mathbf{J}' - \mathbf{J}' = 0. \quad (16)$$

The requirement of Eq. (15) therefore ensures that condition (a) above is satisfied. However, it only partly specifies the vector constants $\mathbf{j}_1^J$, $\mathbf{j}_2^J$, and $\mathbf{l}^J$. Complete specification of these constants is obtained by the requirement that condition (b) be satisfied, as outlined below.

It is convenient to use bra-ket notation in terms of which the transitional-mode kinetic energy T_{rot} [Eq. (8)] is written

$$T_{\text{rot}} = \langle j_1 | \mathbf{T}_1 | j_1 \rangle + \langle j_2 | \mathbf{T}_2 | j_2 \rangle + \langle l | \mathbf{T}_0 | l \rangle, \quad (17)$$

where the kinetic energy tensors $\mathbf{T}_0$, $\mathbf{T}_1$, and $\mathbf{T}_2$ are expressed in terms of the respective principal-axial systems of the orbital and fragment rotors as

$$\mathbf{T}_0 = \begin{pmatrix} B_0 & 0 & 0 \\ 0 & B_0 & 0 \\ 0 & 0 & \infty \end{pmatrix}, \quad (18)$$

$$\mathbf{T}_1 = \begin{pmatrix} C_1 & 0 & 0 \\ 0 & B_1 & 0 \\ 0 & 0 & A_1 \end{pmatrix}, \quad (19)$$

and

$$\mathbf{T}_2 = \begin{pmatrix} C_2 & 0 & 0 \\ 0 & B_2 & 0 \\ 0 & 0 & A_2 \end{pmatrix}. \quad (20)$$

Note that the relationship between the kinetic energy tensors above and the corresponding inertia tensors is established in a straightforward fashion to be

$$\mathbf{T} = \frac{\hbar^2}{2} \mathbf{I}^{-1} \quad (21)$$

since the kinetic energy T of any rotor may be written

$$\begin{aligned} T &= \frac{1}{2} \langle \omega | \mathbf{I} | \omega \rangle \\ &= \frac{\hbar^2}{2} \langle \mathbf{I}^{-1} j | \mathbf{I} | \mathbf{I}^{-1} j \rangle \\ &= \frac{\hbar^2}{2} \langle \mathbf{I}^{-1} j | j \rangle \\ &= \left\langle j \left| \frac{\hbar^2}{2} \mathbf{I}^{-1} \right| j \right\rangle \\ &= \langle j | \mathbf{T} | j \rangle \end{aligned} \quad (22)$$

where $|\omega\rangle$ is the angular velocity vector. Substituting Eq. (14) into Eq. (17), one obtains

$$\begin{aligned} T_{\text{rot}} = & \langle j_1^J | \mathbf{T}_1 | j_1^J \rangle + 2 \langle p_1 | \mathbf{T}_1 | p_1 \rangle \\ & + \langle j_2^J | \mathbf{T}_2 | j_2^J \rangle + 2 \langle p_2 | \mathbf{T}_2 | p_2 \rangle \\ & + \langle J^J | \mathbf{T}_0 | J^J \rangle + 2 \langle p_0 | \mathbf{T}_0 | p_0 \rangle + \langle p_0 | \mathbf{T}_0 | p_0 \rangle, \end{aligned} \quad (23)$$

where the diagonal property of the inertia tensors in the principal-axial frames has been utilized to set $\langle p_1 | \mathbf{T}_1 | j_1^J \rangle = \langle j_1^J | \mathbf{T}_1 | p_1 \rangle$, etc. The cross terms may be rearranged as follows:

$$\langle p_1 | \mathbf{T}_1 | j_1^J \rangle = \left\langle p_1 \left| \frac{\hbar^2}{2} \mathbf{I}_1^{-1} \right| \hbar^{-1} \mathbf{I}_1 \omega_1^* \right\rangle = \frac{\hbar}{2} \langle p_1 | \omega_1^* \rangle, \quad (24)$$

where $|\omega_1^*\rangle$ is the angular velocity vector associated with the angular momentum j_1^J , with $|\omega_2^*\rangle$ and $|\omega_0^*\rangle$ to be similarly defined

$$j_1^J = \mathbf{I}_1 | \omega_1^* \rangle, \quad j_2^J = \mathbf{I}_2 | \omega_2^* \rangle, \quad J^J = \mathbf{I}_0 | \omega_0^* \rangle. \quad (25)$$

Similar rearrangement of the remaining two cross terms in Eq. (23) yields

$$\begin{aligned} T_{\text{rot}} = & \langle j_1^J | \mathbf{T}_1 | j_1^J \rangle + \langle j_2^J | \mathbf{T}_2 | j_2^J \rangle + \langle J^J | \mathbf{T}_0 | J^J \rangle + \langle p_1 | \mathbf{T}_1 | p_1 \rangle \\ & + \langle p_2 | \mathbf{T}_2 | p_2 \rangle + \langle p_0 | \mathbf{T}_0 | p_0 \rangle + \hbar \langle \omega_1^* | p_1 \rangle \\ & + \hbar \langle \omega_2^* | p_2 \rangle + \hbar \langle \omega_0^* | p_0 \rangle. \end{aligned} \quad (26)$$

The means of obtaining a separation of the kinetic energy and a complete specification of the vector constants j_1^J , j_2^J , and J^J now becomes apparent, since if

$$|\omega_1^*\rangle = |\omega_2^*\rangle = |\omega_0^*\rangle = |\omega^*\rangle, \quad (27)$$

then

$$\begin{aligned} \hbar \langle \omega_1^* | p_1 \rangle + \hbar \langle \omega_2^* | p_2 \rangle + \hbar \langle \omega_0^* | p_0 \rangle \\ = \hbar \langle \omega_1^* | p_1' \rangle + \hbar \langle \omega_2^* | p_2' \rangle + \hbar \langle \omega_0^* | p_0' \rangle \\ = \hbar \langle \omega^* | p_1' + p_2' + p_0' \rangle = 0, \end{aligned} \quad (28)$$

where the fact that the momenta $\{p_i\}$ sum to zero has been used [Eq. (16)]. The required value for $|\omega^*\rangle$ is readily determined by substitution of Eqs. (25) and (27) into Eq. (15)

$$\begin{aligned} \mathbf{J}' = & \mathbf{I}_1' | \omega^* \rangle + \mathbf{I}_2' | \omega^* \rangle + \mathbf{I}_0' | \omega^* \rangle \\ = & (\mathbf{I}_1' + \mathbf{I}_2' + \mathbf{I}_0') | \omega^* \rangle \\ = & \mathbf{I}'(\mathbf{q}) | \omega^* \rangle, \end{aligned} \quad (29)$$

or

$$|\omega^*\rangle = |\omega^{J'}\rangle = \mathbf{I}'(\mathbf{q})^{-1} \mathbf{J}, \quad (30)$$

where $\mathbf{I}'(\mathbf{q})$ is the inertia tensor for the overall body of the loose transition state expressed relative to the orbital-rotor principal axes. That is, $|\omega^*\rangle$ is precisely the angular velocity vector $|\omega^{J'}\rangle$ which would generate the angular momentum vector $\mathbf{J}$ by a “rigid body” rotation of the overall system at configuration $\mathbf{q}$. In Eq. (29), the result has been used that the overall inertia tensor $\mathbf{I}'(\mathbf{q})$ of the system at

configuration $\mathbf{q}$ is the sum of the individual inertia tensors $\mathbf{I}_1'$, $\mathbf{I}_2'$, and $\mathbf{I}_0'$ of the fragments and the orbital rotor

$$\mathbf{I}'(\mathbf{q}) = \mathbf{I}_1' + \mathbf{I}_2' + \mathbf{I}_0'. \quad (31)$$

The proof of this relationship is outlined briefly in Appendix A. Note that in Eqs. (29)–(31), the inertia tensors $\mathbf{I}_1'$, $\mathbf{I}_2'$, and $\mathbf{I}_0'$ for the fragments and the orbital rotation are evaluated with reference to their respective centers of mass and are expressed in terms of the common set of Cartesian axes. The common axes have been chosen to be the principal axes of the orbital rotation, hence $\mathbf{I}_0' = \mathbf{I}_0$. The configurational dependence of $\mathbf{I}(\mathbf{q})$ lies in the fact that $\mathbf{I}_1'$ and $\mathbf{I}_2'$ in Eq. (31) are in general nondiagonal, being related to their respective diagonal forms by the orthogonal transformations which rotate their principal axes into orientations parallel to the orbital-rotor principal axes.

Equations (25), (27), and (30) serve to completely specify the vector constants j_1^J , j_2^J , and J^J in terms of their projections onto the common set of Cartesian axes,

$$j_1^{J'} = \mathbf{I}_1' \mathbf{I}'(\mathbf{q})^{-1} \mathbf{J}', \quad j_2^{J'} = \mathbf{I}_2' \mathbf{I}'(\mathbf{q})^{-1} \mathbf{J}', \quad J^{J'} = \mathbf{I}_0' \mathbf{I}'(\mathbf{q})^{-1} \mathbf{J}'. \quad (32)$$

These are simply the angular momenta obtained by decomposing the total angular momentum vector $\mathbf{J}'$ into its contributions from the fragment rotors and the orbital rotation if the overall system at configuration $\mathbf{q}$ were rotating as a rigid body. The analogy between Eqs. (14) and (32) above and the corresponding transformation utilized by Jellinek and Li⁷⁴ for the classical kinetic energy operator of an arbitrary N -body system is clear. Nyman *et al.*⁷⁵ have used the technique to derive an expression for the E - and J -resolved momentum-space volume of an N -body system at a fixed configuration $\mathbf{q}$.

Since the angular momentum vector $\mathbf{J}$ is a constant in the integrals of Eq. (12), the momenta j_1^J , j_2^J , and J^J are, as stated above, vector constants. One therefore has $dj_i = d\mathbf{p}_i$, $d\mathbf{l} = d\mathbf{p}_0$, i.e., the Jacobian $J_c = 1$. The transformation of Eq. (14) is therefore summarized as

$$(j_1, j_2, J) \xrightarrow{J_c=1} (\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_0). \quad (33)$$

The external rotational energy of Eq. (26) denoted E^J is verified by substitution of Eqs. (21) and (32) and subsequent manipulations to be

$$\begin{aligned} E^J = & \langle j_1^J | \mathbf{T}_1 | j_1^J \rangle + \langle j_2^J | \mathbf{T}_2 | j_2^J \rangle + \langle J^J | \mathbf{T}_0 | J^J \rangle \\ = & C(\mathbf{q}) J_x^2 + B(\mathbf{q}) J_y^2 + A(\mathbf{q}) J_z^2, \end{aligned} \quad (34)$$

where J_x , J_y , and J_z are the principal-axis components of the total angular momentum vector $\mathbf{J}$. The delta function $\delta(\mathbf{J} - \mathbf{j})$ is expressed in terms of the new momenta as

$$\begin{aligned} \delta(\mathbf{J} - \mathbf{j}) = & \delta(\mathbf{J}' - \mathbf{j}') \\ = & \delta[\mathbf{J}' - (\mathbf{j}_1' + \mathbf{j}_2' + \mathbf{l}')] \\ = & \delta[\mathbf{J}' - (\mathbf{j}_1^{J'} + \mathbf{j}_2^{J'} + \mathbf{l}^{J'}) - (\mathbf{p}_1' + \mathbf{p}_2' + \mathbf{p}_0')] \\ = & \delta(\mathbf{p}_1' + \mathbf{p}_2' + \mathbf{p}_0'). \end{aligned} \quad (35)$$

The momentum space $\Phi(E, \mathbf{J}, \mathbf{q})$ [Eq. (12)] now becomes

$$\Phi(E, \mathbf{J}, \mathbf{q}) = \frac{1}{\pi^3} \int \cdots \int d\mathbf{p}_1 d\mathbf{p}_2 d\mathbf{p}_0 S[E^* - V(\mathbf{q}) - T_{\text{rot}}] \times \delta(\mathbf{p}'_1 + \mathbf{p}'_2 + \mathbf{p}'_0), \quad (36)$$

where $d\mathbf{p}_1 = dp_{1x} dp_{1y} dp_{1z}$, $d\mathbf{p}_2 = dp_{2x} dp_{2y} dp_{2z}$, $d\mathbf{p}_0 = dp_{0x} dp_{0y} dp_{0z}$ (recall that the unprimed momenta are in the respective principal-axis frames, hence $l_z = l'_z = p_{0z} = 0$), and T_{rot} is obtained by substitution of Eqs. (27), (30), and (34) into Eq. (26) as

$$T_{\text{rot}} = E^J + \hbar \langle \omega^J | p'_1 + p'_2 + p'_0 \rangle + \langle p_1 | \mathbf{T}_1 | p_1 \rangle + \langle p_2 | \mathbf{T}_2 | p_2 \rangle + \langle p_0 | \mathbf{T}_0 | p_0 \rangle. \quad (37)$$

The delta function in Eqs. (35) and (36) has the effect of collapsing three of the eight integrals and eliminating the cross terms between external and internal rotation in Eq. (37). As a result, the kinetic energy is rigorously separated into the external rotational energy E^J and the internal rotational energy T^{int} . The elimination of three of the internal momenta by the delta function will introduce extensive configurational coupling *within* T^{int} , i.e., between the remaining five internal momenta. By virtue of its being a quadratic form, however, the internal kinetic energy is readily diagonalized by an appropriate orthogonal transformation of the momenta with unit Jacobian.⁷⁶ We formally represent the diagonalized form of the internal kinetic energy as

$$T^{\text{int}} = \sum_{i=1}^5 c_i \gamma_i^2, \quad (38)$$

where $\{\gamma_i\}$ are the new momenta in terms of which the internal kinetic energy is separable. The momentum-space volume $\Phi(E, \mathbf{J}, \mathbf{q})$ is written in terms of these momenta as

$$\Phi(E, \mathbf{J}, \mathbf{q}) = \frac{1}{\pi^3} \int d\gamma_1 \cdots d\gamma_5 S \left[E^* - V(\mathbf{q}) - E^J - \sum_{i=1}^5 c_i \gamma_i^2 \right]. \quad (39)$$

Equation (39) is cast readily into the form of Dirichlet's integral²⁹ and evaluated to give

$$\Phi(E, \mathbf{J}, \mathbf{q}) = \frac{1}{\pi^3} \left(\prod_{i=1}^5 c_i^{-1/2} \right) \times \frac{\Gamma(1/2)^5}{\Gamma[(5/2)+1]} [E^* - V(\mathbf{q}) - E^J]^{5/2} \quad (40)$$

where $\Gamma(x)$ is the gamma function. From Eq. (40), it is apparent that one need not determine separately the constants $\{c_i\}$ by diagonalization of the internal kinetic energy T_{int} , since it is simply the *product* of the constants which appears in the expression for the momentum space. This product may be determined as follows: Integration of $\Phi(E, \mathbf{J}, \mathbf{q})$ in Eq. (40) over *all* energetically allowed angular momenta $\mathbf{J}$ yields the momentum space $\Phi(E, \mathbf{q})$ (i.e., summed over $\mathbf{J}$)

$$\Phi(E, \mathbf{q}) = \frac{1}{\pi^3} \frac{1}{[A(\mathbf{q})B(\mathbf{q})C(\mathbf{q})]^{1/2}} \left(\prod_{i=1}^5 c_i^{-1/2} \right) \times \frac{\Gamma(1/2)^8}{\Gamma[(8/2)+1]} [E^* - V(\mathbf{q})]^{8/2} \quad (41)$$

However, as is well known, $\Phi(E, \mathbf{q})$ may also be determined directly from the principal-axis momenta $\{\mathbf{j}_1, \mathbf{j}_2, \mathbf{l}\}$ and the kinetic energy in the form of Eq. (8) (e.g., Ref. 53)

$$\Phi(E, \mathbf{q}) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} d\mathbf{j}_1 d\mathbf{j}_2 d\mathbf{l} S[E - V(\mathbf{q}) - T_{\text{rot}}]. \quad (42)$$

Equation (42) is put into the form of Dirichlet's integral and evaluated to give

$$\Phi(E, \mathbf{q}) = \frac{1}{\pi^3} \frac{1}{B_0} \left(\prod_{i=1}^6 \Lambda_i^{-1/2} \right) \frac{\Gamma(1/2)^8}{\Gamma[(8/2)+1]} [E - V(\mathbf{q})]^{8/2} \quad (43)$$

where $\{\Lambda_i\}$ are the rotational constants of the two asymmetric-top fragments. Comparison of Eqs. (41) and (43) yields the solution

$$\left(\prod_{i=1}^5 c_i^{-1/2} \right) = \frac{[A(\mathbf{q})B(\mathbf{q})C(\mathbf{q})]^{1/2}}{B_0} \left(\prod_{i=1}^6 \Lambda_i^{-1/2} \right) \quad (44)$$

hence

$$\Phi(E, \mathbf{J}, \mathbf{q}) = \frac{1}{\pi^3} \frac{[A(\mathbf{q})B(\mathbf{q})C(\mathbf{q})]^{1/2}}{B_0} \left(\prod_{i=1}^6 \Lambda_i^{-1/2} \right) \times \frac{\Gamma(1/2)^5}{\Gamma[(5/2)+1]} [E^* - V(\mathbf{q}) - E^J]^{5/2}. \quad (45)$$

Equation (45) is the exact analytical expression for the momentum-space volume of an asymmetric top + asymmetric top loose transition state with a given total energy E , total angular momentum vector $\mathbf{J}$, and configuration $\mathbf{q}$. The corresponding result for arbitrary combinations of fragments is

$$\Phi(E, \mathbf{J}, \mathbf{q}) = \frac{1}{\pi} \frac{[A(\mathbf{q})B(\mathbf{q})C(\mathbf{q})]^{1/2}}{B_0} \left(\prod_{i=1}^s \Lambda_i^{-1/2} \right) \times \frac{\Gamma(1/2)^{s-p-1}}{\Gamma[(s-1)/2+1]} \times [E^* - V(\mathbf{q}) - E^J]^{(s-1)/2}, \quad (46)$$

where s is the total number of rotational degrees of freedom of the fragments (e.g., $s=4$ for the linear + linear case; $s=6$ for the nonlinear + nonlinear case), and $p=2$ if one of the fragments is monatomic, otherwise $p=4$. Note that for symmetric fragments, the degenerate rotational constants Λ_i must be repeated the appropriate number of times in the product (e.g., this gives B_i^{-1} if fragment i is linear, or $B_i^{-3/2}$ if fragment i is a spherical top).

The momentum-space volume $\Phi(E, \mathbf{J}, \mathbf{q})$ is obtained by integrating Eq. (46) over all energetically allowed orientations of the angular momentum vector [Eq. (13)]. It is

convenient to make a transformation from the principal-axial components of $\mathbf{j}$ to cylindrical coordinates specifying its magnitude j , its projection K onto the z -principal axis, and the orientation ξ of its projection onto the (x,y) principal-axial plane

$$(j_x, j_y, j_z) \xrightarrow{J_c=j} (j, K, \xi), \quad (47)$$

where

$$j^2 = j_x^2 + j_y^2 + j_z^2, \quad K = j_z, \quad \xi = \tan^{-1} \left(\frac{j_y}{j_x} \right). \quad (48)$$

The momentum space $\Phi(E, J, \mathbf{q})$ is then written

$$\begin{aligned} \frac{\Phi(E, J, \mathbf{q})}{(2J+1)} &= \frac{[A(\mathbf{q})B(\mathbf{q})C(\mathbf{q})]^{1/2}}{B_0} \\ &\times \left(\prod_{i=1}^s \Lambda_i^{-1/2} \right) \frac{\Gamma(1/2)^{s-p-1}}{\Gamma[(s-1)/2 + 1]} \\ &\times \frac{1}{\pi} \int_0^{2\pi} d\xi \int_0^J dK [E - V(\mathbf{q}) - E^J]^{(s-1)/2} \end{aligned} \quad (49)$$

where the integral over j has been evaluated with the use of the delta function $\delta(J-j)$ and the external rotational energy E^J is now

$$E^J = B^*(\mathbf{q}, \xi)J(J+1) + [A(\mathbf{q}) - B^*(\mathbf{q}, \xi)]K^2, \quad (50)$$

in which

$$B^*(\mathbf{q}, \xi) = C(\mathbf{q})\cos^2 \xi + B(\mathbf{q})\sin^2 \xi. \quad (51)$$

Equations (49) and (50) have been "requantized" in the usual semiclassical fashion^{42,77} in order to ensure that the correct behavior is obtained as $J \rightarrow 0$. As indicated in Eq. (49), it is convenient to divide out the $(2J+1)$ degeneracy associated with the projection of the angular momentum vector onto an arbitrary polar axis, since this cancels out in the expression for the rate coefficient.

Evaluation of the integral over K in Eq. (49) yields

$$\begin{aligned} \frac{\Phi(E, J, \mathbf{q})}{(2J+1)} &= \frac{[A(\mathbf{q})B(\mathbf{q})C(\mathbf{q})]^{1/2}}{B_0} \\ &\times \left[\prod_{i=1}^s \left(\frac{1}{\Lambda_i} \right)^{1/2} \right] \frac{\Gamma(1/2)^{s-p}}{\Gamma[(s/2) + 1]} \frac{1}{2\pi} \int_0^{2\pi} d\xi \\ &\times \frac{[E^* - V(\mathbf{q}) - B^*(\mathbf{q}, \xi)J(J+1)]^{s/2}}{[A(\mathbf{q}) - B^*(\mathbf{q}, \xi)]^{1/2}}, \\ &C(\mathbf{q})J(J+1) \leq E^* - V(\mathbf{q}) \leq A(\mathbf{q})J(J+1) \end{aligned} \quad (52)$$

and

$$\begin{aligned} \frac{\Phi(E, J, \mathbf{q})}{(2J+1)} &= \frac{[A(\mathbf{q})B(\mathbf{q})C(\mathbf{q})]^{1/2}}{B_0} \\ &\times \left[\prod_{i=1}^s \left(\frac{1}{\Lambda_i} \right)^{1/2} \right] \frac{\Gamma(1/2)^{s-p-1}}{\Gamma[(s+1)/2]} G(E, J, \xi, \mathbf{q}), \\ &A(\mathbf{q})J(J+1) \leq E^* - V(\mathbf{q}), \end{aligned} \quad (53)$$

where $G(E, J, \xi, \mathbf{q})$ has the following forms for even and odd values of s , respectively:

$$\begin{aligned} G(E, J, \xi, \mathbf{q}) &= \frac{1}{2\pi} \int_0^{2\pi} d\xi \left\{ (2J+1) \left[\sum_{\nu=0}^{s/2-1} \frac{(s-1; -2; \nu)}{(s; -2; \nu+1)} [E^* - V(\mathbf{q}) - B^*(\mathbf{q}, \xi)J(J+1)]^\nu \right. \right. \\ &\times [E^* - V(\mathbf{q}) - A(\mathbf{q})J(J+1)]^{[(s-1)/2] - \nu} \left. \left. + \left(\frac{1}{2} \right)^{(s-2)/2} \frac{(1; 2; s/2)}{(s/2)!} \frac{[E^* - V(\mathbf{q}) - B^*(\mathbf{q}, \xi)J(J+1)]^{s/2}}{[A(\mathbf{q}) - B^*(\mathbf{q}, \xi)]^{1/2}} \right. \right. \\ &\times \left. \left. \sin^{-1} \left[\frac{(2J+1)[A(\mathbf{q}) - B^*(\mathbf{q}, \xi)]^{1/2}}{2[E^* - V(\mathbf{q}) - B^*(\mathbf{q}, \xi)J(J+1)]^{1/2}} \right] \right\}, \quad \text{for } s \text{ even} \end{aligned} \quad (54)$$

and

$$\begin{aligned} G(E, J, \mathbf{q}) &= (2J+1) \frac{1}{2\pi} \int_0^{2\pi} d\xi \left\{ \sum_{\nu=0}^{(s-1)/2} \frac{(s-1; -2; \nu)}{(s; -2; \nu+1)} [E^* - V(\mathbf{q}) - B^*(\mathbf{q}, \xi)J(J+1)]^\nu \right. \\ &\times \left. [E^* - V(\mathbf{q}) - A(\mathbf{q})J(J+1)]^{[(s-1)/2] - \nu} \right\}, \quad \text{for } s \text{ odd.} \end{aligned} \quad (55)$$

The notation $(x; n; \nu)$ in Eqs. (54) and (55) has the following meaning:

$$(x; n; \nu) = \prod_{j=1}^{\nu} [x + (j-1)n]. \quad (56)$$

Note that $(x; n; 0)$ is defined to be unity. Thus, evaluation of the exact momentum-space volume $\Phi(E, J, \mathbf{q})$ requires

only a simple one-dimensional numerical integral over the projection angle ξ .

It is apparent from Eq. (49) that the configuration dependence of the momentum-space volume $\Phi(E, J, \mathbf{q})$ is driven by two factors: (a) the noncentral potential $V(\mathbf{q})$, which alters the total available kinetic energy $E^* - V(\mathbf{q})$; and (b) the external rotational constants $C(\mathbf{q})$, $B(\mathbf{q})$, and most importantly $A(\mathbf{q})$, the relative magnitudes of which

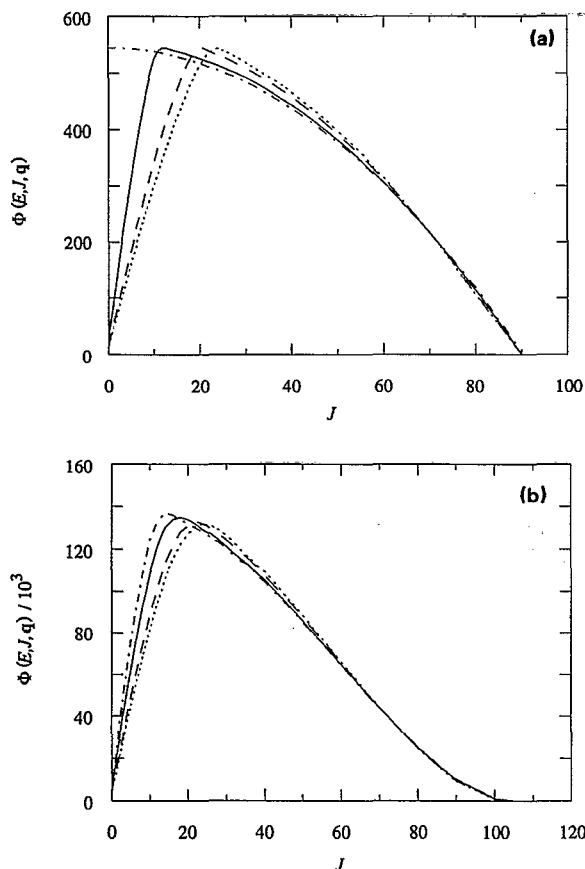


FIG. 1. (a) A plot of available transitional-mode momentum-space volume $\Phi(E, J, \mathbf{q})$ [Eqs. (52) and (53)] vs J for an atom + linear loose transition state, that of NO + O. $E - V(\mathbf{q}) \equiv 1000 \text{ cm}^{-1}$, $R = 3.5 \text{ \AA}$, and $r_{\text{eq}}(\text{NO}) = 1.1508 \text{ \AA}$ (Ref. 42). Each curve is for a different angle θ of the NO fragment relative to the line joining the centers of mass; — ($\theta = 0^\circ$); - - ($\theta = 30^\circ$); - · - ($\theta = 60^\circ$); and --- ($\theta = 90^\circ$). (b) A plot of available transitional-mode momentum-space volume $\Phi(E, J, \mathbf{q})$ [Eqs. (52) and (53)] vs J for a linear + linear loose transition state, that of NC + NO. $E - V(\mathbf{q}) \equiv 1000 \text{ cm}^{-1}$, $R = 3.5 \text{ \AA}$, and $r_{\text{eq}}(\text{NC}) = 1.1718 \text{ \AA}$ (Ref. 37). Each curve is for a different angle θ of the NO fragment relative to the line joining the centers of mass; - - - ($\theta = 0^\circ$); — ($\theta = 30^\circ$); - · - ($\theta = 60^\circ$); and --- ($\theta = 90^\circ$). The angle of the NC fragment relative to the line joining the centers of mass is fixed at 45° in each case. The torsional angle between the fragments is 90° in each case.

determine to what extent the “ K ” rotor is able to act as an “internal” degree of freedom (i.e., constrained only by energetic and not angular-momentum boundary conditions). Figure 1 illustrates the J dependence and the configuration

($\mathbf{q}$) dependence of the momentum space for typical atom + linear (NO + O) and linear + linear (NC + NO) fragment combinations. In each case, the energy $E^* - V(\mathbf{q})$ has been set to a constant value of 1000 cm^{-1} for all curves (i.e., all configurations) in order to focus on the kinematic effects involved. The atom + linear and linear + linear combinations have been chosen because they exhibit the most extreme variations in $A(\mathbf{q})$ with changing configuration, and hence the most extreme configurational dependence due to factor (b) above. The center-of-mass separation R in each case is taken as $3.5 \text{ \AA}$. It should be noted that in the evaluation of the sum of states, the noncentral potential $V(\mathbf{q})$ will cause configurations of minimum potential to be most favored through the weighting factor $[E - V(\mathbf{q})]^{(s-1)/2}$ in Eq. (49).

A common feature of loose transition states is that they are highly prolate, i.e., for any configuration $\mathbf{q}$, one has $B(\mathbf{q}) \approx C(\mathbf{q})$. The momentum space $\Phi(E, J, \mathbf{q})$ of Eqs. (52)–(55) may be evaluated analytically if the overall body of the loose transition state is a symmetric (oblate or prolate) top. Thus, if the approximation is made that $B(\mathbf{q}) = C(\mathbf{q})$ (i.e., a prolate top), one has from Eq. (51) that $B^*(\mathbf{q}, \zeta) = B(\mathbf{q})$, leading directly to

$$\frac{\Phi(E, J, \mathbf{q})}{(2J+1)} = \frac{B(\mathbf{q})[A(\mathbf{q})]^{1/2}}{B_0[A(\mathbf{q}) - B(\mathbf{q})]^{1/2}} \left[\prod_{i=1}^s \left(\frac{1}{\Lambda_i} \right)^{1/2} \right] \times \frac{\Gamma(1/2)^{s-p}}{\Gamma[(s/2)+1]} [E^* - V(\mathbf{q}) - B(\mathbf{q})J(J+1)]^{s/2},$$

$$B(\mathbf{q})J(J+1) \leq E^* - V(\mathbf{q}) \leq A(\mathbf{q})J(J+1) \quad (57)$$

and

$$\frac{\Phi(E, J, \mathbf{q})}{(2J+1)} = \frac{B(\mathbf{q})[A(\mathbf{q})]^{1/2}}{B_0} \left[\prod_{i=1}^s \left(\frac{1}{\Lambda_i} \right)^{1/2} \right] \times \frac{\Gamma(1/2)^{s-p-1}}{\Gamma[(s+1)/2]} G(E, J, \mathbf{q}),$$

$$A(\mathbf{q})J(J+1) \leq E^* - V(\mathbf{q}). \quad (58)$$

where $G(E, J, \mathbf{q})$ has the following forms for even and odd values of s , respectively:

$$G(E, J, \mathbf{q}) = (2J+1) \left\{ \sum_{\nu=0}^{s/2-1} \frac{(s-1; -2; \nu)}{(s; -2; \nu+1)} [E^* - V(\mathbf{q}) - B(\mathbf{q})J(J+1)]^\nu [E^* - V(\mathbf{q}) - A(\mathbf{q})J(J+1)]^{[(s-1)/2]-\nu} \right. \\ \left. + \left(\frac{1}{2} \right)^{(s-2)/2} \frac{(1; 2; s/2)}{(s/2)!} \frac{[E^* - V(\mathbf{q}) - B(\mathbf{q})J(J+1)]^{s/2}}{[A(\mathbf{q}) - B(\mathbf{q})]^{1/2}} \sin^{-1} \left[\frac{(2J+1)[A(\mathbf{q}) - B(\mathbf{q})]^{1/2}}{2[E^* - V(\mathbf{q}) - B(\mathbf{q})J(J+1)]^{1/2}} \right] \right\}, \quad \text{for } s \text{ even} \quad (59)$$

and

$$G(E, J, \mathbf{q}) = (2J+1) \left\{ \sum_{\nu=0}^{(s-1)/2} \frac{(s-1; -2; \nu)}{(s; -2, \nu+1)} [E^* - V(\mathbf{q}) - B(\mathbf{q})J(J+1)]^\nu \right. \\ \left. \times [E^* - V(\mathbf{q}) - A(\mathbf{q})J(J+1)]^{[(s-1)/2] - \nu} \right\}, \text{ for } s \text{ odd.} \quad (60)$$

Equations (57)–(60) have been presented in our earlier work.⁶³ As noted in the Introduction, these equations were originally derived with reference to a specific set of “easy” configurations in which the angular coupling in the kinetic-energy expression disappears. The general functional form of Eqs. (57) and (58) was identified in the results for each of the specific configurations and this functional form was proposed as an accurate means of interpolating between these configurational limits by allowing $A(\mathbf{q})$ and $B(\mathbf{q})$ to vary continuously with changing configuration. The development presented in this work shows that Eqs. (57) and (58) are in fact no interpolation, but rather a *general result for any configuration* $\mathbf{q}$ of a loose transition state. The only approximation involved is that the overall body of the loose transition state at the configuration $\mathbf{q}$ may be taken as a prolate top. The prolate approximation is found to work exceptionally well for the typical examples presented in Fig. 1; the analytic results of Eqs. (57) and (58) agreeing with the exact numerical evaluation to within 1% for all of the curves presented.

III. CONCLUSION

Accurate and rapid calculation of microscopic and thermal rate coefficients for reactions involving potential surfaces which have no barrier to recombination is an important objective because of the ubiquitous nature of such reactions and because of the experimental and theoretical challenges they offer to our understanding of reaction rate theory. The calculation of such rate coefficients is also a very difficult objective, the main problems being associated with the energy and angular momentum dependence of the transition state or channel maxima, the complexity of the noncentral potential, and the complexity of the angular momentum coupling between separating or recombining fragments.

In this work, we have shown that the exact evaluation of the classical sum of states for the transitional modes (i.e., the fragment and orbital rotations) may be simplified dramatically by the analytical evaluation of the available momentum-space volume $\Phi(E, J, \mathbf{q})$ for the loose transition state at a given configuration $\mathbf{q}$ with a total system energy E and angular momentum J . The result capitalizes on the development of a technique by Jellinek and Li⁷⁴ for separating instantaneously the kinetic energy of a system into external rotational energy and internal kinetic energy. The present work proves the validity for arbitrary configurations $\mathbf{q}$ of the expressions for the momentum space which were proposed in our previous work⁶³ on the basis of a less complete analysis. The development of rapid and accurate

calculational techniques towards achieving the above objective is thus greatly facilitated.

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APPENDIX A: PROOF OF ADDITIVITY OF FRAGMENT AND ORBITAL INERTIA TENSORS

The result of Eq. (31) that the inertia tensor for the overall body of the loose transition state is given by the sum of the inertia tensors for the fragments and the orbital rotor is now proven. Let $\{\mathbf{r}_i, i=1, N\}$ be the position vectors of all atoms in the system relative to the overall center of mass. Fragment 1 will include atoms $1, 2, \dots, p$, the remaining $N-p$ atoms constituting fragment 2. The overall mass of the system is

$$M = \sum_{i=1}^N m_i \quad (A1)$$

The masses of fragments 1 and 2 are, respectively,

$$M_1 = \sum_{i=1}^p m_i, \quad M_2 = \sum_{i=p+1}^N m_i \quad (A2)$$

and the position vectors for their centers of mass are $\mathbf{R}_1$ and $\mathbf{R}_2$. The position vector of atom i relative to the center of mass of the fragment to which it belongs will be denoted $\mathbf{r}'_i$, such that

$$\sum_{i=1}^p m_i \mathbf{r}'_i = \sum_{i=p+1}^N m_i \mathbf{r}'_i = 0. \quad (A3)$$

Note that the position vectors $\mathbf{r}_i$ and $\mathbf{r}'_i$ are related by

$$\mathbf{r}_i = \mathbf{R}_1 + \mathbf{r}'_i, \quad i=1, \dots, p, \\ \mathbf{r}_i = \mathbf{R}_2 + \mathbf{r}'_i, \quad i=p+1, \dots, N. \quad (A4)$$

The inertia tensor is written in terms of dyadic notation (e.g., Ref. 78) as

$$\mathbf{I}(\mathbf{q}) = \sum_{i=1}^N m_i (r_i^2 \mathbf{1} - \mathbf{r}_i \mathbf{r}_i) = \sum_{i=1}^N m_i [(\mathbf{r}_i \mathbf{r}_i) \mathbf{1} - \mathbf{r}_i \mathbf{r}_i], \quad (A5)$$

where $\mathbf{1}$ is the unit dyadic. Substituting Eq. (A4) into Eq. (A5),

$$\begin{aligned} \mathbf{I}(\mathbf{q}) = & \sum_{i=1}^p m_i [(\mathbf{R}_1 + \mathbf{r}'_i) \cdot (\mathbf{R}_1 + \mathbf{r}'_i) \mathbf{1} - (\mathbf{R}_1 + \mathbf{r}'_i)(\mathbf{R}_1 + \mathbf{r}'_i)] \\ & + \sum_{i=p+1}^N m_i [(\mathbf{R}_2 + \mathbf{r}'_i) \cdot (\mathbf{R}_2 + \mathbf{r}'_i) \mathbf{1} \\ & - (\mathbf{R}_2 + \mathbf{r}'_i)(\mathbf{R}_2 + \mathbf{r}'_i)]. \end{aligned} \quad (\text{A6})$$

Expanding the scalar products and dyads, and eliminating cross terms with the use of Eq. (A3), one obtains the desired result

$$\begin{aligned} \mathbf{I}(\mathbf{q}) = & \sum_{i=1}^2 M_i (\mathbf{R}_i^2 \mathbf{1} - \mathbf{R}_i \mathbf{R}_i) + \sum_{i=1}^p m_i (r_i'^2 \mathbf{1} - \mathbf{r}'_i \mathbf{r}'_i) \\ & + \sum_{i=p+1}^N m_i (r_i'^2 \mathbf{1} - \mathbf{r}'_i \mathbf{r}'_i) \\ = & \mathbf{I}'_0 + \mathbf{I}'_1 + \mathbf{I}'_2, \end{aligned} \quad (\text{A7})$$

where the inertia tensors $\mathbf{I}_0$, $\mathbf{I}_1$, and $\mathbf{I}_2$ are denoted with a prime in Eq. (A7) because they are expressed in terms of a common cartesian axial system, chosen in the text to be the principal axes of the orbital rotor.

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